RUDRAEDGE AEROSPACE

Project - Rudra Edge RE-T1(Spectra)
Daily Engineering Log

1. TEAM MEMBER INFORMATION

Engineer Name: Suraj Nadda
Designation / Role: Software & Data Engineer
Task Order Reference: 13 October 2026 – Auditing and Logging Framework

2. DAILY LOG ENTRY FORM

- **Date:** 13 October 2026

- **1. Assigned Task:**

  System Auditing, Structured JSON Logging, and Error Trace Capturing.

- **2. What Was Studied:**

  Studied the concepts of application-level system auditing and structured logging. Learned how timestamps, event types, actions, status, and details are stored in JSON format. Studied Python's logging module, exception handling, and traceback capturing. Also understood how JSON Lines file can be validated and used to review application events.

- **3. What Was Implemented:**

  Developed and executed a standalone Python audit-log program. Implemented structured event recording, timestamp generation, and controlled error trace capturing. Added JSON validation and generated an execution summary containing event counts and error information. The program completed successfully, recording six events in the log and capturing one controlled test error.

- **4.Tools / Libraries Used:**

  VS Code, PowerShell, Python 3.14.6, CSV , Pathlib , JSON.

- **5.Output File Name-** suraj_rudra_task/ Day9/codes/audit_system.py

  Path-https://github.com/surajnadda29/suraj_rudra_t1.git

- **6. Problems Encountered:**
  No problem was encountered

- **7. How Problem Was Resolved:**
  No problem was encountered.

- **8. Next Task Scheduled:**

  (14 October 2026)- Performance benchmarking : loop vs vectorized throughput analysis.

- **9. Personal Expenditure: ₹0 (Verified Mandate)**

3. ENGINEER SELF-DECLARATION & SIGN-OFF

I hereby formally declare that today's technical progression, synthetic dataset code, and local repository paths have been documented truthfully in compliance with Section 6 (Common Technical Standards). I certify with absolute transparency that my personal financial expenditure for today's assigned task is exactly ₹0, utilizing strictly local open-source infrastructure.

Candidate/Engineer Signature:  Date: 13/10/2026

Full Name: Suraj Nadda

PROJECT LEAD VERIFICATION BOX — TO BE FILLED BY SARVESH SHARMA ONLY

I have reviewed today's technical output, local repository commits, and verified the project compliance standards for this day's assigned task.

Task Evaluation Status:

[ ] Approved / Completed

[ ] Needs Revision / Delayed

[ ] Technical Block Encountered

Operational Expenditure Confirmed: ₹0 (Strict Zero-Cost Compliant)

Remarks/Notes: _____

Project Lead Signature: _____ Date: ____/____/2026